

Problem I Consider the following optimization problem,

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - y\|_2^2 + \lambda R_{\alpha, \beta}(x) + i_{\mathbb{R}^n_+}(x)$$

where, $R_{\alpha, \beta}(x) = \|x\|_1 + \langle \alpha, x \rangle + \beta$, $\alpha \in \mathbb{R}^n$, $\beta > 0$
and A is convolution operator, $Ax = h * x$

① Derive the Proximal operator of the regularization term $R_{\alpha, \beta}$.

→ To derive the Proximal operator of the regularization term, $R_{\alpha, \beta}(x) = \|x\|_1 + \langle \alpha, x \rangle + \beta$, we need the definition of Proximal operator,

$$\text{prox}_{\lambda R_{\alpha, \beta}}(y) = \underset{x \in \mathbb{R}^n}{\operatorname{argmin}} \left\{ \frac{1}{2} \|x - y\|_2^2 + \lambda R_{\alpha, \beta}(x) \right\}$$

$$= \underset{x \in \mathbb{R}^n}{\operatorname{argmin}} \left\{ \frac{1}{2} \|x - y\|_2^2 + \lambda \|x\|_1 + \lambda \langle \alpha, x \rangle + \lambda \beta \right\}$$

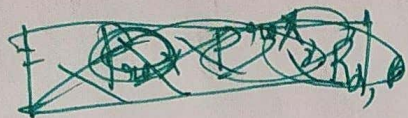
The term $\lambda \beta$ is constant with respect to x , so it does not affect the minimizer and can be omitted.

$$\Rightarrow \underset{x \in \mathbb{R}^n}{\operatorname{argmin}} \left\{ \frac{1}{2} \|x - y\|_2^2 + \lambda \|x\|_1 + \lambda \langle \alpha, x \rangle \right\}$$

$$= \underset{x \in \mathbb{R}^n}{\operatorname{argmin}} \left\{ \frac{1}{2} \|x - y\|_2^2 + \lambda \|x\|_1 + \langle \lambda \alpha, x - y \rangle + \frac{1}{2} \|\lambda \alpha\|_2^2 \right\}$$

$$= \operatorname{argmin} \left\{ \frac{1}{2} \|x - (y - \lambda \alpha)\|_2^2 + \lambda \|x\|_1 \right\} \quad (2)$$

$$= \operatorname{prox}_{\lambda \|\cdot\|_1} (y - \lambda \alpha)$$



$\Rightarrow$ As, Soft-Thresholding function, S_λ :

$$S_\lambda(z)_i = \operatorname{sign}(z_i) \cdot \max(|z_i| - \lambda, 0)$$

$$\Rightarrow \operatorname{prox}_{\lambda R_{\alpha, \beta}}(y) = S_{\lambda \beta}(y - \lambda \alpha)$$

~~Shifted soft thresholding function~~
Shifted soft thresholding function.

(2) Give the Pseudo-code for one iteration of FISTA.

$$\Rightarrow \text{Objective Func} \Rightarrow \min_{x \in \mathbb{R}^n} \frac{1}{2} \|Ax - y\|_2^2 + \lambda R_{\alpha, \beta}(x) + i_{\gamma_0}(x);$$

Algorithm $\Rightarrow$ FISTA Iteration for Convolution Based Sparse Reconstruction.

Required input $\Rightarrow$ Previous iteration: $x_k, x_{k-1} \in \mathbb{R}^n$.

• Momentum parameter: $t_k \in \mathbb{R}$

• Step size: $\gamma \in \mathbb{R}^+$

• Regularization parameters: $\lambda, \alpha \in \mathbb{R}^n, \beta \in \mathbb{R}^+$

• Convolution kernel: $h \in \mathbb{R}^n$

• Observed data: $y \in \mathbb{R}^n$

Outputs:-

① Updated iterates: x_{k+1}, t_{k+1}

Primal Gdch

③

① Update Momentum Parameter $\Rightarrow t_{k+1} \leftarrow \frac{1 + \sqrt{1 + 4t_k^2}}{2}$

② Compute Extrapolated Point y_k $\Rightarrow y_k \leftarrow x_k + \left(\frac{t_k - 1}{t_{k+1}} \right) (x_k - x_{k-1})$

③ Gradient Step in Fourier Domain $\Rightarrow$

$$\text{FFT}(A y_k) \leftarrow \text{FFT}(R) \odot \text{FFT}(y_k)$$

$$\text{FFT}(x_k) \leftarrow \text{FFT}(A y_k) - \text{FFT}(y)$$

$$\nabla b(y_k) \leftarrow \text{IDFT}(\text{FFT}(R)^* \odot \text{FFT}(y_k))$$

$$x_{k+1/2} \leftarrow y_k - \gamma \nabla b(y_k)$$

④ Proximal Step for $\lambda R_{\alpha, \beta}(x)$ $\Rightarrow$

$$x_{k+1}^{\text{temp}} \leftarrow S_{\gamma \lambda} (x_{k+1/2} - \gamma \lambda d)$$

where, $S_{\gamma}(z)_i = \text{sign}(z_i) \cdot \max(|z_i| - \gamma, 0)$.

⑤ Project on Non Negative Constraint $\Rightarrow$

~~$$x_{k+1} = \max(0, S_{\gamma \lambda} (x_{k+1/2} - \gamma \lambda d))$$~~

$$x_{k+1} \leftarrow \max(0, x_{k+1}^{\text{temp}})$$

③ Analyse the role of the parameter β in the reconstruction quality of the solution.

④

$\Rightarrow \beta$ does not influence reconstruction quality; it is an irrelevant additive constant.